# Transfinite induction as least-counterexample existence: any formula instantiated at some ordinal is instantiated at an epsilon-minimal ordinal

The Open Logic Project contributors

openlogic:set-theory/ordinals:local-snapshot-20261013; CC BY 4.0

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-set-theory-ordinals/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The counted proof is the author's complete argument: both branches of the case split and the single named citation appear verbatim; the author's commented-out precursor variants remain as source comments and deliver no text.

*Editorial note.* The label ordinductionschema names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

*Editorial note.* The counted statement is carried by the author's own thm environment in the unit source.

*Allowed standard prerequisites (editorial).*

- Elements of ordinals are ordinals. Same unit, \ollabel{ordmemberord} (source line 416). Last line of the proof (line 461): the in-least element found inside the ordinal alpha is itself an ordinal.
- Ordinals are well-ordered by membership. The unit's standing facts on ordinals developed just above (section header line 443 states the induction schema and trichotomy are the reason ordinals are well-ordered by membership). Supplies the existence of an in-least phi-element inside alpha in the Otherwise branch.

*Source reference.* Original reference ordinductionschema1: sources/olp-set-theory-ordinals/pinned-original-source.tex, source lines 456–456. Named citation invoked at source line(s) 456: the author's marker \olref{ordinductionschema1} resolves inside the full Open Logic Project to another unit (not staged in this corpus); the cited fact is used as a standard result and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original reference ordinductionschema2: sources/olp-set-theory-ordinals/pinned-original-source.tex, source lines 461–461. Named citation invoked at source line(s) 461: the author's marker \olref{ordinductionschema2} resolves inside the full Open Logic Project to another unit (not staged in this corpus); the cited fact is used as a standard result and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original reference ordmemberord: sources/olp-set-theory-ordinals/pinned-original-source.tex, source lines 416–416. Prerequisite pointer invoked at proof line(s) 460; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

**Theorem 0.1** (Transfinite Induction)**.** *For any formula $\phi(x)$:*

$$\textit{if } \exists\alpha\phi(\alpha), \textit{ then } \exists\alpha(\phi(\alpha) \wedge (\forall\beta \in \alpha)\neg\phi(\beta))$$

*where the displayed quantifiers are implicitly restricted to ordinals.*

*Proof.* Suppose $\phi(\alpha)$, for some ordinal $\alpha$. If $(\forall\beta \in \alpha)\neg\phi(\beta)$, then we are done. Otherwise, as $\alpha$ is an ordinal, it has some $\in$-least !!element which is $\phi$, and this is an ordinal by ordmemberord. □